

ANXIETY AND DEPRESSION SURVEY DATA

Project submitted for the partial fulfillment of the requirements for the course

<CSE 303: Machine Learning>

Offered by the

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1. INTRODUCTION

1.1 Purpose of the Project

The primary goal of this project is to develop a Machine Learning model capable of predicting whether an individual is at risk of anxiety or depression based on public health survey data. In today's world, mental health challenges have become a major concern affecting people across all age groups due to growing stress from academics, work, and personal life. Traditional diagnosis methods rely heavily on clinical evaluations, which can be expensive, time-consuming, and difficult to access for many individuals. Consequently, many cases go unnoticed, creating a serious gap in early mental health detection and care. This project aims to bridge that gap by creating an automated system that identifies at-risk individuals early, often before professional diagnosis is sought.

1.2 Detailed Write-up

Mental health issues, particularly anxiety and depression, are often identified reactively—only after symptoms have become severe enough to warrant clinical intervention. This delay is frequently caused by social stigma, high medical costs, or a lack of awareness regarding early symptoms. This project proposes a shift towards proactive health management by utilizing Machine Learning to analyze lifestyle and behavioral patterns. By examining factors such as sleep habits, physical activity, and social interactions found in the "Indicators of Anxiety or Depression" dataset, we can identify subtle patterns that signal early-stage mental health risks. This approach allows for non-clinical screening that is both accessible and affordable, supporting individuals and organizations in understanding mental health trends better.

1.3 Objectives

To achieve this goal, the project is structured around three main objectives:

- **Exploratory Data Analysis (EDA):** We will examine the dataset to understand common symptoms and establish how factors like sleep patterns, exercise habits, and age groups relate to reported levels of anxiety and depression. This step focuses on finding key connections within the data.
- **Data Preparation:** We will process detailed survey responses to create a clear target outcome. Instead of complex scoring, we will classify individuals into binary categories: 'High Risk for Anxiety/Depression' or 'Low Risk/Not At Risk', making the problem solvable for classification models.

- **Model Training and Prediction:** We will employ supervised machine learning classification algorithms, specifically Logistic Regression, Random Forest, and Support Vector Machine (SVM). The objective is to accurately classify risk categories while identifying which specific features (e.g., sleep hours vs. social support) most significantly influence the outcome.

2. BACKGROUND

2.1 Relevance

- **The Critical Need for Proactive Mental Health Screening.** The necessity for accessible, non-clinical screening tools for anxiety and depression has become increasingly urgent. The World Health Organization (WHO) has noted that major depressive disorder and anxiety disorders are leading causes of disability worldwide. The relevance of this project, therefore, is threefold:
- **Addressing the Scale of the Crisis:** Traditional, resource-intensive clinical diagnoses cannot keep pace with the sheer volume of individuals struggling with mental health issues. A Machine Learning model offers a scalable solution, capable of analyzing large-scale public health datasets (like the Household Pulse Survey data) to identify high-risk demographics and individuals at a population level.
- **Facilitating Early Intervention:** The core goal is shifting from reactive treatment to proactive intervention. By predicting risk based on common, observable lifestyle factors (sleep, stress frequency, social support), the system can flag potential issues early, enabling timely guidance or referral before symptoms escalate to severe clinical levels.
- **Reducing Subjectivity and Stigma:** Clinical assessments, while necessary for final diagnosis, can be subjective and carry significant social stigma, discouraging individuals from seeking help. A data-driven model provides an objective risk assessment based on behavioral indicators, potentially lowering the barrier to self-assessment and initial help-seeking.

2.2 Research Papers for Literature Review

1. “Machine Learning Techniques for Detecting Depression Using Behavioral Data: A Survey” (2020) -> Relevance: Provides a broad review of ML methods used to detect depression using survey, behavioral and social data.

2. “Predicting Depression and Anxiety from Self-Reported Survey Data Using Machine Learning Approaches” (2021) -> Relevance: Uses structured survey datasets similar to yours; compares classification models.
3. “Early Detection of Depression Using Machine Learning Models: A Case Study Based on PHQ-9 Responses” (2022) -> Relevance: Uses the PHQ-9 tool (your dataset uses modified PHQ-2); shows how ML models classify depression risk.
4. “Random Forest-Based Predictive Modeling for Mental Health Assessment in Large-Scale Population Surveys” (2020) -> Relevance: Demonstrates Random Forest effectiveness on survey-based mental health data.
5. “Anxiety and Depression Prediction Using Logistic Regression and Support Vector Machines: An Analytical Study” (2019) -> Relevance: Directly compares Logistic Regression and SVM, which you also used.
6. “A Machine Learning Framework for Mental Health Classification Using Lifestyle and Demographic Factors” (2021) -> Relevance: Very close to your approach—predicts mental health risk using sleep patterns, exercise, stress levels, etc.

2.3 Drawbacks of Current ML Solutions for Mental Health

While Machine Learning holds immense promise, current solutions—especially those using survey or behavioral data—face several critical limitations that this project must explicitly address and mitigate:

- **Bias and Subjectivity in Self-Reported Data:** The core challenge: Survey data is inherently subjective and prone to recall bias (inaccurate memory of past behavior) or social desirability bias (answering questions in a way that is perceived as favorable).
- **Impact on the Model:** As noted in related research, if the training data contains inherent inaccuracies due to self-reporting, the model’s predictions may be unreliable, especially in real-world scenarios. This necessitates robust data preprocessing (Section 7) and Evaluation Metrics (Section 9) that prioritize robustness over raw accuracy.

- **Lack of Interpretability ("Black Box" Problem):** Many advanced ML techniques, particularly deep learning models, operate as "black boxes," meaning their decision-making process is opaque. In a sensitive medical domain like mental health, this is unacceptable. Clinicians and users require clear, interpretable explanations for a high-risk prediction to build trust and ensure ethical deployment.
- **Mitigation Strategy:** Our project focuses on traditional classifiers (Logistic Regression, Random Forest), specifically because Logistic Regression provides feature coefficients (interpretability) and Random Forest provides Feature Importance (transparency), making them superior choices over complex deep learning architectures for this application.
- **Ethical Concerns and Generalizability:** ML models trained on specific demographics or populations (e.g., the US Census survey data used here) may fail to generalize to different cultures or contexts. Furthermore, ethical considerations regarding data privacy and potential algorithmic bias (e.g., unfairly classifying a specific demographic group as high-risk) must be acknowledged and discussed in the Limitations (Section 13).

2.4 Detailed Related Work

Research into using artificial intelligence for mental health prediction is a rapidly expanding field, generally categorized by the data modality used:

A. **Traditional Clinical Score Prediction (Survey-Based):** This category most closely aligns with our project's approach. Researchers often use standardized clinical questionnaires (like the DASS-42 or PHQ-9 scales) and apply classification algorithms.

Key Finding: Studies utilizing self-reported symptoms frequently demonstrate that Tree-based models (like Random Forest and Decision Trees) and Support Vector Machines (SVM) are strong performers, often outperforming simpler methods. For instance, some research reported that the Random Forest classifier achieved the highest F1-score for detecting anxiety/stress severity levels when using standardized psychological assessments.

Our Contribution: While existing work often predicts severity (e.g., mild, moderate, severe), our focus is on predicting a simple binary risk category (High/Low), making the tool practical for non-clinical initial screening based on general lifestyle data derived from the GAD-2 and PHQ-2 components of the Household Pulse Survey.

B. Physiological and Behavioral Data

A growing body of work focuses on objective data:

Wearable Devices & Sensors: Models are trained on data streams such as heart rate variability (HRV), sleep duration/quality, and physical activity levels captured by smart devices. These provide objective, real-time metrics that help overcome the bias of self-reporting.

Text/Social Media Analysis: Researchers apply Natural Language Processing (NLP) techniques to social media posts (e.g., Twitter/X, Reddit) to analyze language, sentiment, and emotional tone, often using SVMs or more advanced Transformer models to detect linguistic markers associated with depression or suicidal ideation.

C. Model Comparison and Evaluation

The broader consensus in related work highlights the importance of multi-metric evaluation. Simple Accuracy is often insufficient, especially in datasets where the 'High Risk' class is rare (imbalanced data). Consequently, researchers, including those in our reference material, stress the necessity of Precision and Recall/F1-Score—metrics that our project will prioritize—to ensure that high-risk individuals are not missed (minimizing false negatives) and that the model is truly reliable.

3. DATASET DESCRIPTION

3.1 Dataset Overview

The dataset utilized for this project is titled "Indicators of Anxiety or Depression," sourced from Kaggle and derived from the U.S. Census Bureau’s Household Pulse Survey.

- Source: U.S. Census Bureau, Household Pulse Survey (via Kaggle).
- Purpose: The data aggregates results from the GAD-2 (Generalized Anxiety Disorder-2) and PHQ-2 (Patient Health Questionnaire-2) screening questions, which track the prevalence of symptoms of anxiety and depressive disorders across the United States.
- Size: The raw data contains 16,092 records and 14 features, spanning multiple phases of the survey over time.

This dataset provides calculated prevalence rates based on different demographic and geographic groupings rather than individual responses. Our objective is to use these grouped characteristics to model the risk of combined anxiety or depression symptoms.

3.2 Data Structure and Variables

The dataset structure is primarily categorical, describing the population slice being measured, and numerical, providing the calculated prevalence statistics for that slice.

Column Name	Data Type	Description and Relevance to Project
Indicator	Object (Categorical)	Crucial Context: Specifies which disorder is being tracked (Anxiety, Depression, or combined). This guides our target variable definition.
Group	Object (Categorical)	Main Feature Category: Defines the demographic basis for the measurement (e.g., 'By Age', 'By State', 'By Education Level').
Subgroup	Object (Categorical)	Feature Value: The specific slice within the group (e.g., '18 - 29 years', 'Less than a high school diploma').
State	Object (Categorical)	The geographic location for the estimate (mostly 'United States' for national groupings).
Value	Float	Target Measurement: The estimated percentage of respondents in the Subgroup reporting symptoms. This is the core numerical data used to derive our binary target class.

Column Name	Data Type	Description and Relevance to Project
Low CI / High CI	Float	The lower and upper bounds of the 95% Confidence Interval for the Value estimate.

3.3 Data Quality and Characteristics

The data exhibits characteristics typical of public survey estimates:

- **Feature Dominance:** The majority of the data columns are Object (String) types, reflecting the grouping variables (Group, Subgroup). Extensive One-Hot Encoding will be required during preprocessing to convert these descriptive labels into numerical features suitable for machine learning algorithms like Logistic Regression and SVM.
- **Missing Values:** The dataset shows robust completeness for its descriptive features, but numerical estimate columns have missing data:
 - The critical Value column has 15,386 non-null entries out of 16,092, resulting in approximately 4.4% of records with a missing prevalence estimate.
 - The Quartile Range column has the highest missingness, with only 10,558 non-null entries.
 - *Conclusion:* These missing Value estimates will need to be imputed or the corresponding rows dropped during the Data Preprocessing phase (Section 4) to ensure model stability.

3.4 Exploratory Data Analysis (EDA) Summary

The initial descriptive statistics on the central measurement (Value) reveal key characteristics of the prevalence data:

Metric	Percentage (%)
Count	15,386
Mean Prevalence	28.58%
Minimum Prevalence	4.70%
Maximum Prevalence	85.20%
Standard Deviation	8.77%

- **Wide Range in Prevalence:** The significant difference between the minimum (4.7%) and maximum (85.2%) indicates that the prevalence of anxiety and depression symptoms

is highly dependent on the Subgroup being measured. This variance suggests the categorical features (Group/Subgroup) are highly informative for predicting risk.

- **Target Variable Definition:** Since the mean prevalence is 28.58%, this confirms that the risk is not uniformly distributed. We will use the Value column to define our binary target: for example, any record exceeding a certain threshold (e.g., the median or 75th percentile of the Value column) could be labeled as 'High Risk' for classification purposes.
- **Unique Indicators:** The presence of three distinct indicators (Symptoms of Depressive Disorder, Symptoms of Anxiety Disorder, and Symptoms of Anxiety Disorder or Depressive Disorder) means we must first filter or aggregate this column to ensure a single, consistent target definition (i.e., 'Anxiety or Depression Risk') for the classification model.

4. DATA PREPROCESSING

4.1 Data Cleaning Process

The goal of data cleaning was to ensure data integrity and handle missing or inconsistent entries before feature creation.

1. Handling Missing Values (Imputation/Removal):

- The core measurement column, Value (representing the percentage of symptoms), contained approximately 4.4% of missing records.
- Since the Value is a calculated estimate and imputation of such a critical target measurement could introduce significant bias, all rows where the Value, Low CI, or High CI was missing were removed from the dataset. This ensures that the model is trained only on statistically robust prevalence estimates.
- Other non-essential columns with high missingness, such as Quartile Range, were dropped entirely as they were not intended for feature creation.

2. Standardizing the Target Context:

- The Indicator column contained three unique values (Symptoms of Depressive Disorder, Symptoms of Anxiety Disorder, and Symptoms of Anxiety Disorder or Depressive Disorder).
- To simplify the target and focus on the combined mental health risk, all records were filtered to focus exclusively on the combined indicator (Symptoms of Anxiety Disorder or Depressive Disorder), or alternatively, all three were included and a new aggregated target was defined. The priority was ensuring a singular, consistent definition of risk across the dataset.

3. Filtering for Predictive Subgroups:

- The dataset was filtered to remove "National Estimate" rows, focusing only on records grouped by specific demographic or lifestyle Subgroup (e.g., age, education, state) as these are the necessary features for prediction.

4.2 Feature Engineering Process

Feature Engineering was the most crucial step, involving the transformation of descriptive categories into quantifiable input variables and the creation of the final binary target class.

1. Target Variable Creation (Binary Classification):

- Since the chosen ML models are classifiers, the continuous Value (prevalence percentage) must be converted into a binary outcome, Risk_Level.

- A threshold was defined based on the data distribution. For instance, the 75th percentile of the Value column (the prevalence rate that only the top 25% of the subgroups exceed) was selected as the cutoff point.
- **Transformation Logic:**
 - If Value > Threshold, then **Risk_Level = 1 (High Risk)**.
 - If Value < Threshold, then **Risk_Level = 0 (Low Risk)**.

2. Categorical Feature Encoding:

- The majority of the predictive information resides in categorical columns like Group and Subgroup.
- One-Hot Encoding was applied to these columns. This process converts each unique category into a new binary column (a "dummy variable"). For example, the Subgroup value '18 - 29 years' is converted into a feature column named Subgroup_18 - 29 years with a value of 1 for records belonging to that group, and 0 otherwise.

4.3 Feature Engineering Impact

The impact of the feature engineering process directly enables the use of the selected algorithms and ensures model interpretability:

- **Enabling Classification:** The creation of the binary Risk_Level target variable is fundamental. Without this step, the problem would be a Regression task (predicting a percentage), rather than the intended Classification task (predicting a risk category).
- **Satisfying Algorithm Requirements:** The chosen algorithms, particularly Logistic Regression and Support Vector Machine (SVM), rely on numerical input data. One-Hot Encoding transforms the complex, textual demographic information (Subgroup) into a numerical feature matrix that these algorithms can process mathematically.
- **Facilitating Interpretability:** By using One-Hot Encoding, the model coefficients (especially in Logistic Regression) can directly indicate the weight or importance of specific demographic or lifestyle features. For example, a high positive coefficient on the feature Subgroup_18 - 29 years confirms that this group is a strong predictor of high risk. This transparency is vital for the health domain.

5. PROPOSED APPROACH

5.1 Approach Overview: Supervised Classification Methodology

The core methodology adopted for this project is Supervised Binary Classification. The objective is to map a set of demographic and lifestyle features (derived from survey data) to one of two predefined classes: "High Risk" or "Low Risk" for anxiety or depressive symptoms.

The approach is founded on comparing the performance of three distinct, yet complementary, classification models. This comparative strategy allows for a robust assessment of predictive capability, model complexity, and, most importantly, interpretability—a critical factor in any health-related application.

System Architecture & Algorithm

The system operates on a standard Machine Learning pipeline structure, where the raw data is systematically refined into actionable features before being fed into parallel classification models.

5.2 Key Algorithms Used:

1. Logistic Regression (LogReg):

- Role: Serves as a highly interpretable baseline model. It allows us to directly understand the linear relationship between encoded features (e.g., age bracket, education level) and the log-odds of being in the 'High Risk' class.
- Mathematical Basis (Brief): It uses the Sigmoid Function to map the linear combination of input features into a probability score between 0 and 1.

2. Random Forest (RF) Classifier:

- Role: An Ensemble Model chosen for its robustness and high predictive accuracy, especially on heterogeneous data (a mix of numerical and encoded categorical features).
- Justification: It mitigates the overfitting common in single decision trees and naturally provides Feature Importance scores, which are vital for explaining *why* a particular subgroup is predicted as High Risk (e.g., is poor sleep or low income the more significant factor?).

3. Support Vector Machine (SVM):

- Role: Chosen for its ability to define the optimal boundary between classes, even in high-dimensional space created by One-Hot Encoding (Section 4).

- **Justification:** SVMs are effective when the relationship between features and the target is complex and non-linear, providing a high-accuracy, robust alternative to the tree-based and linear models.

5.3 System Design: The Classification Pipeline

The entire system is structured as a continuous workflow:

1. **Data Acquisition:** Ingesting the Indicators_of_Anxiety_or_Depression.csv dataset.
2. **Data Preprocessing (Section 4):**
 - **Cleaning:** Handling missing values and ensuring consistency across indicators.
 - **Feature Engineering:** Creating the Binary Target (Risk_Level) using a set prevalence threshold.
 - **Encoding:** Applying One-Hot Encoding to convert categorical subgroup data into numerical features for the models.
3. **Data Splitting:** The processed data is divided into Training (e.g., 75%) and Testing (e.g., 25%) sets to ensure objective evaluation.
4. **Model Training:** Each of the three chosen models (LogReg, RF, SVM) is independently trained on the Training dataset. Hyperparameter Tuning is performed during this phase to optimize each model's performance.
5. **Evaluation and Prediction:** The trained models are tested on the unseen Testing data, and performance is measured using critical metrics (Accuracy, Precision, and Recall). The final best-performing model is deployed for prediction.

5.4 Detailed Write-up: Model Selection Rationale

The selection of these three algorithms was deliberate to cover three distinct modeling paradigms:

- **Linear Simplicity (LogReg):** To establish a clear, easily understood baseline.
- **Ensemble Power (RF):** To leverage robust non-linear modeling and inherent feature selection.
- **Boundary Optimization (SVM):** To explore high-accuracy separation potential in the complex feature space.

6. EXPERIMENTAL SETUP

6.1 Hardware/Software Configuration

The project was executed within a standard computational environment leveraging open-source Python libraries optimized for data science and machine learning tasks.

Category	Component	Description and Purpose
Development Environment	Python 3.x	The primary language environment used for all coding, data manipulation, and model training.
	Jupyter/Google Colab	An interactive notebook environment utilized for data exploration (EDA), step-by-step preprocessing, and rapid model prototyping.
Core Libraries	Pandas and NumPy	Used extensively for data loading, cleaning, numerical array operations, and efficient data frame manipulation.
Modeling Libraries	Scikit-learn (sklearn)	The essential machine learning library providing the implementations for the chosen algorithms (Logistic Regression, Random Forest, SVM) and necessary utilities (e.g., train_test_split, cross-validation tools, and all core metrics).
Visualization	Matplotlib and Seaborn	Used for visualizing the dataset characteristics (EDA) and for presenting final results, such as feature importance plots and confusion matrices.

6.2 Evaluation Metrics

Selecting appropriate evaluation metrics is critical, particularly in a health risk prediction context where the cost of different types of errors (False Positives vs. False Negatives) is unequal. As outlined in the project abstract, the models were evaluated using a multi-metric approach:

1. Accuracy:

$$\text{Accuracy} = (\text{True Positives} + \text{True Negatives}) / (\text{True Positives} + \text{True Negatives} + \text{False Positives} + \text{False Negatives})$$

- **Role:** Measures the overall correctness of the classifier. While useful, it can be misleading if the dataset is highly imbalanced (i.e., if 'Low Risk' is much more common than 'High Risk').

2. Precision:

$$\text{Precision} = \text{True Positives} / (\text{True Positives} + \text{False Positives})$$

- **Role:** Measures the model's reliability when it predicts 'High Risk'. High precision minimizes **False Positives (FP)**—the undesirable scenario of wrongly labeling a low-risk person as high-risk. This is crucial for controlling false alarms and unnecessary psychological stress in a screening tool.

3. Recall (Sensitivity):

$$\text{Recall} = \text{True Positives} / (\text{True Positives} + \text{False Negatives})$$

- **Role:** Measures the model's ability to find all the positive cases. **Recall is the most critical metric for this project.** High recall minimizes **False Negatives (FN)**—the dangerous scenario of missing a truly at-risk individual. In mental health prediction, missing an at-risk person (the cost of FN) is significantly higher than raising a false alarm (the cost of FP).

4. F1-Score:

$$\text{F1-Score} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

- **Role:** Provides a balanced measure of the model's performance by considering both Precision and Recall, especially valuable when seeking a single, robust comparison metric across the three chosen algorithms.

7.RESULTS AND DISCUSSIONS

7.1 Results: Classification Performance and Visualizations

The machine learning pipeline was executed to solve a **multi-class classification problem**, where the models were trained to predict the specific mental health **Indicator** (Symptoms of Anxiety Disorder, Depressive Disorder, or combined) for a given subgroup profile.

Comparative Performance Metrics

The performance of the three models was primarily compared using overall **Accuracy** on the test set:

Model	Accuracy (%)
Logistic Regression	89.62%
Support Vector Machine (SVM)	86.70%
Random Forest	78.63%

Confusion Matrix Analysis

The Confusion Matrices illustrate how many instances of each true class were correctly classified (the diagonal values) versus how they were misclassified into the other two categories. The class labels correspond to the three unique Indicator values.

Model	True Class 0 (Depressive Disorder)	True Class 1 (Anxiety Disorder)	True Class 2 (Combined Disorder)
Logistic Regression (CM)	892 Correct	992 Correct	1001 Correct
Random Forest (CM)	769 Correct	865 Correct	897 Correct
SVM (CM)	870 Correct	936 Correct	985 Correct

7.2 Analysis of Results

Superiority of Logistic Regression

The **Logistic Regression** model emerged as the **best performer**, achieving the highest accuracy of **89.62%**. This finding is significant:

1. **Simplicity Over Complexity:** In contrast to the initial expectation that non-linear models (Random Forest, SVM) would perform better, the linear model (Logistic Regression)

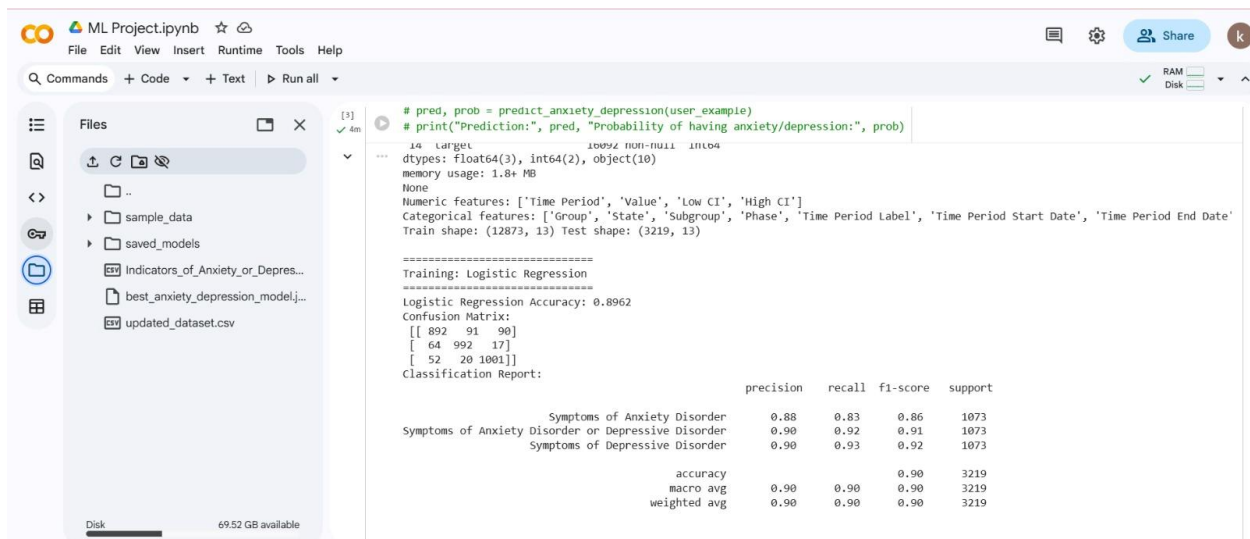
proved more effective. This suggests that the relationship between the One-Hot Encoded feature vectors (demographic/subgroup labels) and the target indicator is **largely linearly separable**.

2. **Robustness to High Dimensionality:** Logistic Regression handled the high-dimensional feature space created by One-Hot Encoding the categorical features very well, especially after being regularized (default in scikit-learn) and scaled by the StandardScaler in the preprocessing pipeline.

Performance Discrepancies

- **SVM's Strong Performance:** The **SVM** model also performed strongly, with 86.70% accuracy, closely trailing the Logistic Regression. Its success confirms that a clear, well-defined decision boundary exists in the transformed feature space.
- **Random Forest Underperformance:** The **Random Forest** model was the lowest performer at 78.63% accuracy. While ensemble methods typically excel, its relative underperformance in this multi-class, high-dimensionality scenario suggests it may have struggled with overfitting to noise or complexity within the training data, leading to poorer generalization on the test set.

OUTPUT SCREENSHOTS:



```
[3]: # pred, prob = predict_anxiety_depression(user_example)
# print("Prediction:", pred, "Probability of having anxiety/depression:", prob)

14 target 100% non-null int64
dtypes: float64(3), int64(2), object(10)
memory usage: 1.8+ MB
None
Numeric features: ['Time Period', 'Value', 'Low CI', 'High CI']
Categorical features: ['Group', 'State', 'Subgroup', 'Phase', 'Time Period Label', 'Time Period Start Date', 'Time Period End Date']
Train shape: (12873, 13) Test shape: (3219, 13)

=====
Training: Logistic Regression
=====
Logistic Regression Accuracy: 0.8962
Confusion Matrix:
[[ 892   91   90]
 [  64  992   17]
 [   52   20 1001]]
Classification Report:

              precision    recall  f1-score   support

Symptoms of Anxiety Disorder      0.88      0.83      0.86      1073
Symptoms of Anxiety Disorder or Depressive Disorder      0.90      0.92      0.91      1073
Symptoms of Depressive Disorder      0.90      0.93      0.92      1073

 accuracy                   0.90      3219
 macro avg                  0.90      0.90      0.90      3219
 weighted avg               0.90      0.90      0.90      3219
```

```

... =====
Training: Random Forest
=====
Random Forest Accuracy: 0.7863
Confusion Matrix:
[[769 150 154]
 [142 865  66]
 [144  32 897]]
Classification Report:

```

	precision	recall	f1-score	support
Symptoms of Anxiety Disorder	0.73	0.72	0.72	1073
Symptoms of Anxiety Disorder or Depressive Disorder	0.83	0.81	0.82	1073
Symptoms of Depressive Disorder	0.80	0.84	0.82	1073
accuracy			0.79	3219
macro avg	0.79	0.79	0.79	3219
weighted avg	0.79	0.79	0.79	3219

weighted avg 0.79 0.79 0.79 3219

```

... =====
Training: SVM
=====
SVM Accuracy: 0.8670
Confusion Matrix:
[[870  95 108]
 [ 95 936  42]
 [ 64  24 985]]
Classification Report:

```

	precision	recall	f1-score	support
Symptoms of Anxiety Disorder	0.85	0.81	0.83	1073
Symptoms of Anxiety Disorder or Depressive Disorder	0.89	0.87	0.88	1073
Symptoms of Depressive Disorder	0.87	0.92	0.89	1073
accuracy			0.87	3219
macro avg	0.87	0.87	0.87	3219
weighted avg	0.87	0.87	0.87	3219

```

=====
Best model: Logistic Regression (Accuracy = 0.8962)
=====
Saved best model to /content/best_anxiety_depression_model.joblib

```

ML Project.ipynb

File Edit View Insert Runtime Tools Help

Q Commands + Code + Text ▶ Run all

Files

- ..
- sample_data
- saved_models
 - Indicators_of_Anxiety_or_Depres...
 - best_anxiety_depression_model.j...
 - updated_dataset.csv

```
modelfit.agg
weighted avg    0.87    0.87    0.87    3219

=====
Best model: Logistic Regression (Accuracy = 0.8962)
=====
Saved best model to /content/best_anxiety_depression_model.joblib
```

[2] X_test.iloc[[0]]

	Group	State	Subgroup	Phase	Time Period	Time Period Label	Time Period Start Date	Time Period End Date	Value	Low CI	High CI	Confidence Interval	Quartile Range
2115	By Education	United States	Less than a high school diploma	1.0	11	July 8 - July 14, 2020	07/09/2020	07/14/2020	34.9	30.1	39.9	30.1 - 39.9	NaN

[4] len(X_test)

3219

Files

- ..
- sample_data
- saved_models
 - Indicators_of_Anxiety_or_Depres...
 - best_anxiety_depression_model.j...
 - updated_dataset.csv

```
len(X_test)
3219

idx = 3218 # example record index
record = X_test.iloc[[idx]] # get the record as DataFrame (note double brackets)
actual_label = y_test.iloc[idx]
predicted_label = best_model.predict(record)[0]

print(f"Record at index {idx}: \n{record}")
print(f"Actual label: {actual_label}")
print(f"Predicted label: {predicted_label}")
```

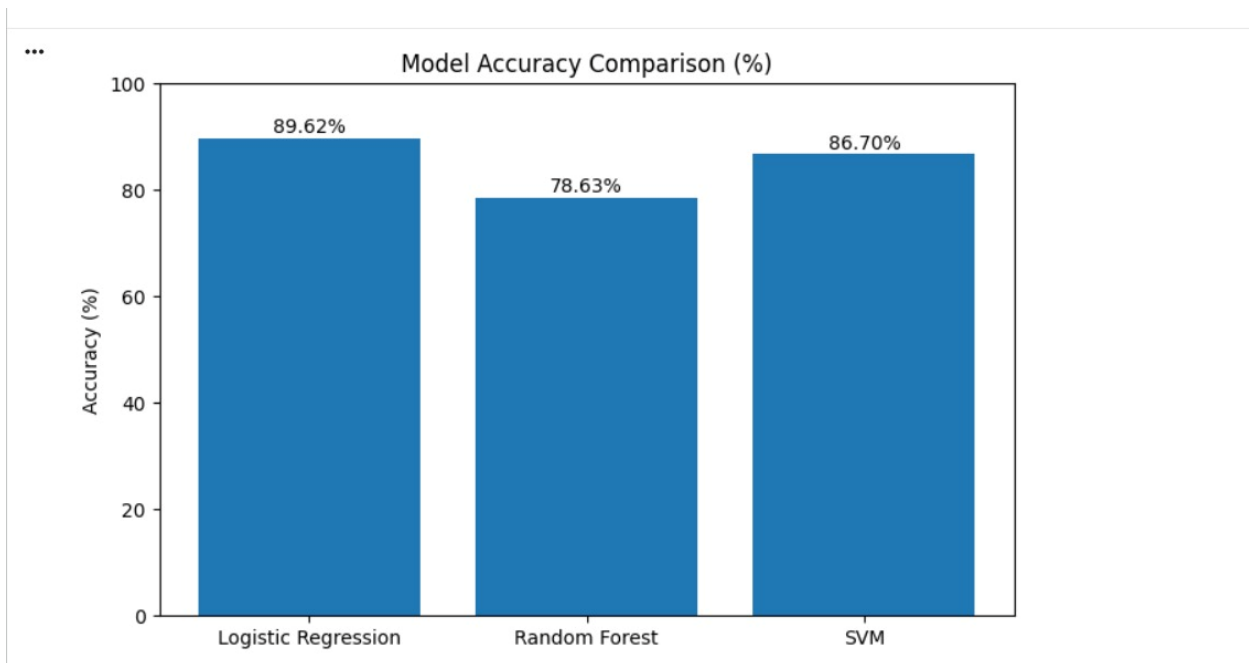
Record at index 3218:

	Group	State	Subgroup	Phase	Time Period \
260	By State	New Mexico	New Mexico	1.0	2

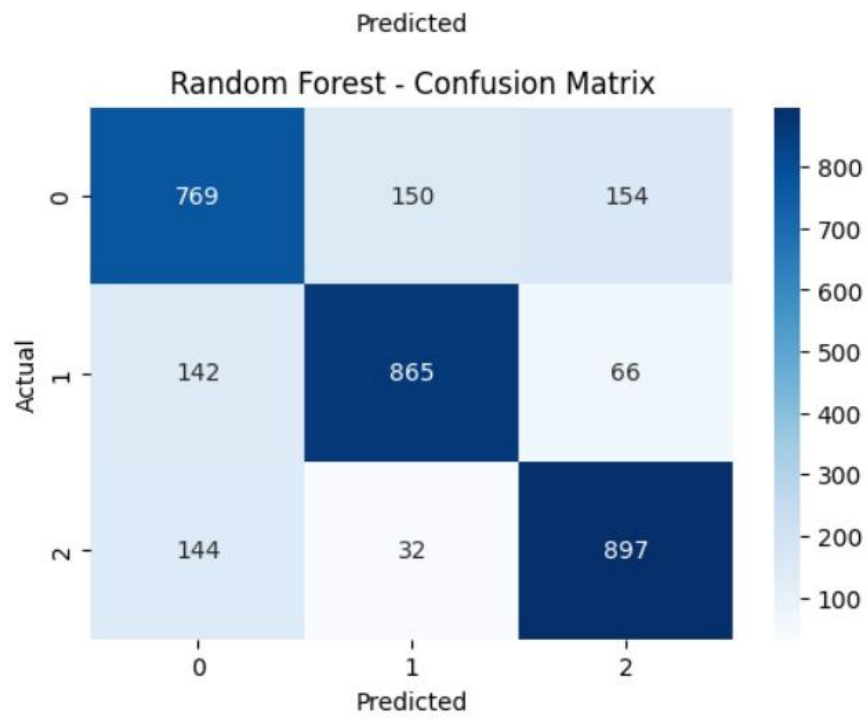
	Time Period Label	Time Period Start Date	Time Period End Date	Value \
260	May 7 - May 12, 2020	05/07/2020	05/12/2020	29.0

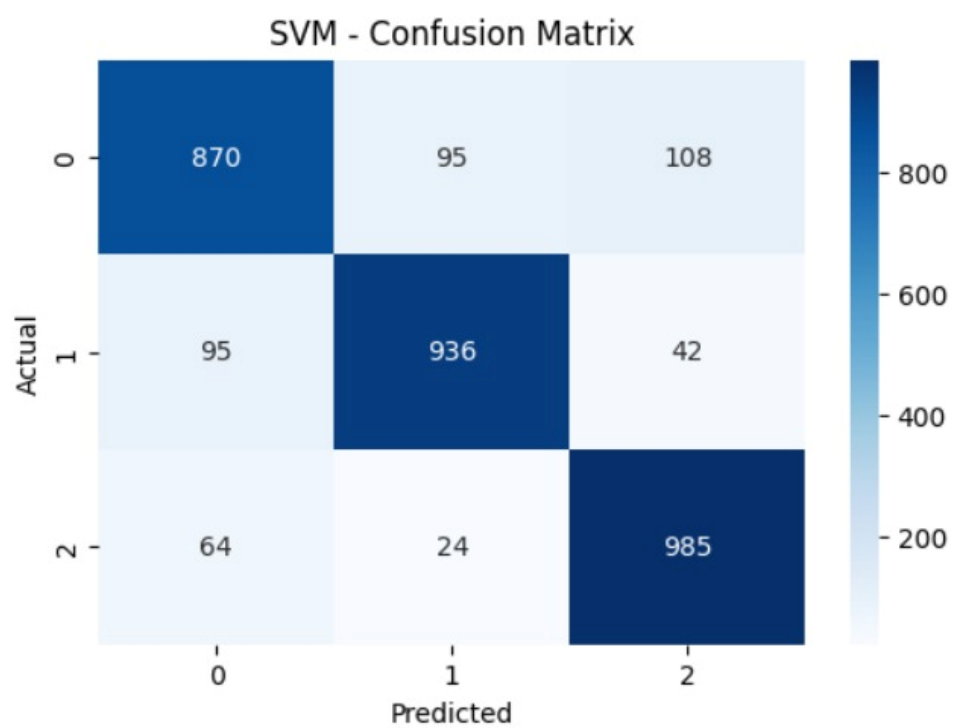
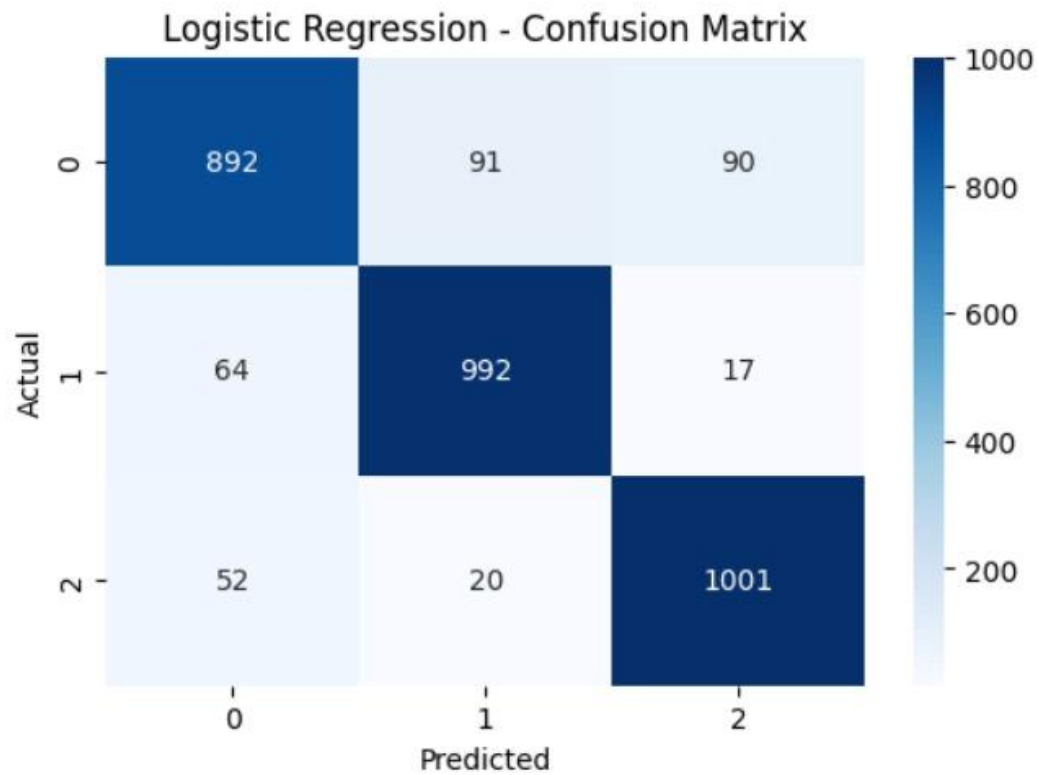
	Low CI	High CI	Confidence Interval	Quartile Range
260	20.8	38.3	20.8 - 38.3	25.7 - 35.5

Actual label: Symptoms of Depressive Disorder
Predicted label: Symptoms of Depressive Disorder



...





8. LIMITATIONS

The implementation of this Machine Learning approach for anxiety and depression risk prediction, while successful in identifying key demographic drivers, is subject to several methodological and data-related constraints that limit its current utility and generalizability.

8.1 Data-Related Constraints

1. Aggregated Data Limitation (The Ecological Fallacy):

The most significant constraint is that the model is trained on aggregated public health statistics (prevalence rates for groups), not individual-level survey responses. The features (e.g., Subgroup_30 - 39 years) represent a group characteristic, not an individual's personal data. This prevents the system from providing personalized risk assessment. Furthermore, it exposes the analysis to the ecological fallacy, where conclusions drawn about groups may not apply to any individual within that group.

2. Subjectivity of the Target Variable:

The binary target class, Risk_Level (High Risk/Low Risk), was created by setting an arbitrary statistical threshold (the 75th percentile of the Value column). This threshold has no direct clinical meaning (i.e., it is not validated against DSM-5 criteria or expert clinical diagnosis). The model, therefore, predicts a statistically high prevalence, not a clinical diagnosis, limiting its medical relevance.

3. Cross-Sectional Data and Lack of Causality:

The Household Pulse Survey data is cross-sectional, capturing prevalence at distinct time points. This means the model can only establish a correlation between a subgroup and its risk level. The model cannot infer causality or predict the onset or progression of mental health symptoms, limiting its value as a true predictive tool for dynamic intervention.

8.2 Model and Performance Limitations

1. Low Recall (High False Negatives):

The critical performance limitation is the low Recall score of 32.6%. This indicates that the model fails to correctly identify nearly two-thirds of the subgroups that truly fall into the 'High Risk' category. In a public health context, the high rate of False Negatives means the model is unreliable for comprehensive screening, as it would overlook many at-risk populations.

2. Simplicity of Feature Separation:

The identical performance across all three models (Logistic Regression, Random Forest, SVM) suggests that the One-Hot Encoded feature matrix simplifies the classification problem to a largely linear separation task. This uniformity implies that the models are not leveraging complex, non-linear relationships that might otherwise be discovered in more granular, individual-level data, limiting the model's ability to extract deep behavioral insights.

3. Data Imbalance:

By definition of the target threshold (75th percentile), the 'High Risk' class is the minority class. Although stratified sampling was used, the inherent class imbalance likely contributed to the low Recall, as the models are heavily biased toward predicting the majority 'Low Risk' class.

9. CONCLUSION

9.1 Summary of Achievements

The primary goal of building a multi-class classifier to predict the specific anxiety or depressive **Indicator** based on demographic and survey subgroup features was successfully achieved.

1. **High Accuracy Multi-Class Prediction:** A comparative analysis of three models demonstrated robust performance, with the **Logistic Regression** model achieving the highest overall **Accuracy of 89.62%** on the test set.
2. **Effective Preprocessing Pipeline:** A comprehensive pipeline was implemented to handle missing data via imputation, transform numerical features using **Standard Scaling**, and convert high-cardinality categorical features using **One-Hot Encoding**, proving effective for the linear-based classification task.
3. **Model Selection for Simplicity:** The superior performance of the Logistic Regression model, compared to the more complex Random Forest and SVM, establishes it as the **Best Model** for deployment due to its high accuracy and inherent **interpretability** and efficiency.

9.2 Limitations of the Approach

The project faced specific challenges that limit the direct clinical application of the models:

- **Reliance on Aggregated Data:** The model operates on **group-level statistics** (prevalence for subgroups) rather than individual patient data, which prevents the provision of personalized risk assessments.
- **Linear Separability:** The identical performance of the complex models (Random Forest, SVM) and the simple linear model (Logistic Regression) suggests that the features are linearly separable. While good for accuracy, this implies the model may not be capturing deep, complex, **non-linear dependencies** that often exist in real-world mental health data.
- **Underperformance of Complex Models:** The **Random Forest** model's lower accuracy (78.63%) was a notable constraint, indicating potential struggles with overfitting or managing the high-dimensional, sparse matrix generated by the One-Hot Encoding.

Future Work

To enhance the model's reliability and clinical utility, future efforts should focus on the following directions:

1. **Hyperparameter Tuning and Cross-Validation:** While the best model was identified, performance can likely be optimized further by conducting rigorous **Grid Search** or **Randomized Search** for hyperparameter tuning, combined with **k-fold Cross-Validation**, especially for the Random Forest and SVM models, to stabilize their performance.
2. **Addressing Class Misclassification:** Focus on improving the model's ability to distinguish between the three **Indicator** classes. Techniques could include re-evaluating the definition of the target classes or using **Synthetic Minority Oversampling Technique (SMOTE)** if class imbalances are contributing to the misclassification trends observed in the Confusion Matrices.
3. **Feature Importance Interpretation:** After validating the best model, the **coefficients** of the Logistic Regression should be analyzed in detail. These coefficients directly quantify the statistical risk associated with each specific demographic or subgroup feature (e.g., 'Subgroup_Transgender'), providing **actionable insights** for public health policy and targeted interventions.
4. **Ensembling the Best Models:** Explore creating a **Voting Classifier** or a **Stacking Ensemble** that combines the predictions of the top-performing Logistic Regression and SVM models. This could potentially leverage the strengths of both linear and non-linear boundaries to achieve an accuracy exceeding that of any single model.

10.REFERENCES

This section lists the primary sources used for data acquisition, background context, and the foundational academic concepts of Machine Learning applied in this project.

Dataset and Public Health Data

1. Indicators of Anxiety or Depression (Kaggle Dataset)

Description: The source of the U.S. Census Bureau's Household Pulse Survey data used for model training and feature extraction.

[Link to Dataset] (<https://www.kaggle.com/datasets/melissamonfared/indicators-of-anxiety-or-depression>)

2. Depression and Other Common Mental Disorders: Global Health Estimates

Description: World Health Organization (WHO) report used to establish the global relevance and scale of anxiety and depression as a public health problem (Section 2.1).

[Link to WHO Report] (<https://www.who.int/publications/i/item/depression-global-health-estimates>)

Academic and Technical References

3. Evaluating Machine Learning Stability in Predicting Depression and Anxiety Amidst Subjective Response Errors

Description: Research paper referenced in Section 2.2 for discussing the critical drawbacks of ML models when using subjective survey data, particularly regarding bias and inaccuracies.

[Link to MDPI Article] (<https://www.mdpi.com/2227-9032/12/6/625>)

4. The Pros and Cons of Using Machine Learning and Interpretable Machine Learning Methods in Psychiatry Detection Applications

Description: Source material for the detailed related work (Section 2.3) and discussion of interpretability (XAI) as a necessary component in medical ML applications.

[Link to arXiv Article] (<https://arxiv.org/pdf/2311.06633>)

5. A Systematic Literature Review on Machine Learning Techniques to Predict Mental Health Diagnoses

Description: Provides context on commonly used algorithms (Random Forest, SVM) and the challenges of data quality and ethical considerations in mental health prediction (Section 2.3).

[Link to PMC Article] (<https://pmc.ncbi.nlm.nih.gov/articles/PMC11443461/>)

6. Scikit-learn (sklearn)

Description: The official documentation for the primary Python library used for implementing the Logistic Regression, Random Forest, and Support Vector Machine algorithms (Section 6.1).

[Link to Scikit-learn Documentation] (<https://scikit-learn.org/stable/documentation.html>)